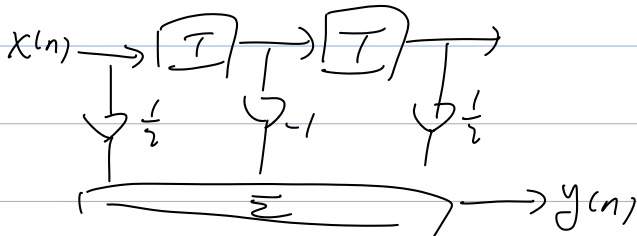


FIR
(i)

$$H(z) = \frac{1}{2} - z^{-1} + \frac{1}{2} z^{-2} = \frac{Y(z)}{X(z)}$$

$$\Rightarrow Y(z) = \frac{1}{2} X(z) - X(z) \cdot z^{-1} + X(z) \cdot \frac{1}{2} z^{-2}$$

$$\Rightarrow y(n) = \frac{1}{2} x(n) - x(n-1] + \frac{1}{2} x(n-2)$$



(ii)

$$y(0) = \frac{1}{2} - 0 + 0 = 0.5$$

$$y(1) = \frac{1}{2} - 1 + 0 = -0.5$$

$$y(2) = \frac{1}{2} - 1 + \frac{1}{2} = 0$$

...

$$y(n) = 0.$$

High-pass filter.

(iii) Let $z = e^{j\omega}$, then:

$$H(e^{j\omega}) = \frac{1}{2} - e^{-j\omega} + \frac{1}{2} \cdot e^{-j2\omega}$$

$$= \frac{1}{2} (1 + e^{-j2\omega}) - e^{-j\omega}$$

$$= \frac{1}{2} e^{-j\omega} (e^{j\omega} + e^{-j\omega}) - e^{-j\omega}$$

$$= e^{-j\omega} (\cos \omega - 1) = -2e^{-j\omega} \sin^2 \frac{\omega}{2}$$

$$\text{or } |H(\omega)| = 2 \sin^2 \frac{\omega}{2}$$

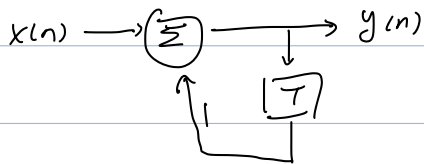
Q4. IIR.

$$H(z) = \frac{1}{1-z^{-1}}$$

$$(a) \quad H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1-z^{-1}}$$

$$\Rightarrow Y(z) - Y(z) \cdot z^{-1} = X(z)$$

$$\Rightarrow y(n) - y(n-1] = x(n)$$



$$(b) \quad \text{From } y(n] = x(n] + y(n-1]$$

$$\text{when } x(n] = \delta(n],$$

$$y(0] = 1 + 0 = 1$$

$$y(1] = 0 + 1 = 1$$

$$y(2] = 0 + 1 = 1$$

...

$$y(n] = 1 \quad (n \geq 0)$$

therefore $y(n] = h(n] = u(n]$ unit step response

$$(c) \quad H(e^{j\omega}) = \frac{1}{1-e^{-j\omega}} \quad (\text{Let } z = e^{j\omega})$$

$$|H(e^{j\omega})| = \frac{1}{|1-e^{-j\omega}|} = \frac{1}{|1-(\cos\omega - j\sin\omega)|}$$

$$= \frac{1}{\sqrt{(1-\cos\omega)^2 + \sin^2\omega}} = \frac{1}{2|\sin\frac{\omega}{2}|}$$

$$\text{therefore, } |H(e^{j\omega})| = \frac{1}{2|\sin\frac{\omega}{2}|}$$

$$\text{when } \omega = 0, \quad |H(e^{j\omega})| \rightarrow \infty$$

when $\omega = \pi$, $|H(e^{j\omega})| = \frac{1}{2}$

Q1. Fourier Transform

(a). $f_s = 8 \text{ kHz}$

$T = 1 \text{ s}$

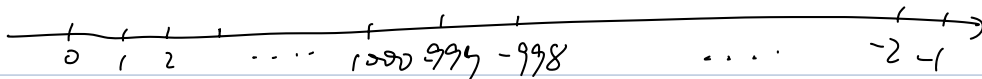
$N = f_s \cdot T = 8000$

DFT bin spacing: $\Delta f = \frac{f_s}{N} = 1 \text{ Hz}$

so DFT bin k corresponds to:

• pos freq: $f_k = k \Delta f$

• neg freq: $f_k = -(N-k) \Delta f$



We want to remove all freq. $|f| < 100 \text{ Hz}$

which means:

$k = 0, 1, \dots, 99$, and $k = N-99, N-98, \dots, N-1$

should be removed.

$$Y[k] = \begin{cases} 0 & , k = 0, 1, \dots, 99, \text{ and } k = N-99, \dots, N-1 \\ X[k] & , \text{ otherwise.} \end{cases}$$

(b). non-linearity.

(i) A non-linearity in the time domain generates new freq. comp. in the spectrum.

produce harmonics of existing tones and intermodulation products.

↓
energy is spread

(ii). Non-linearity generates extra harmonics, especially at higher freq.

↓

increases the spectral brightness and raises the spectral centroid, to which our hearing is very sensitive.

(iv). in non-linearity, louder parts should be driven harder into non-linearity than quiet parts.

if normalize → compress dynamic range

↓

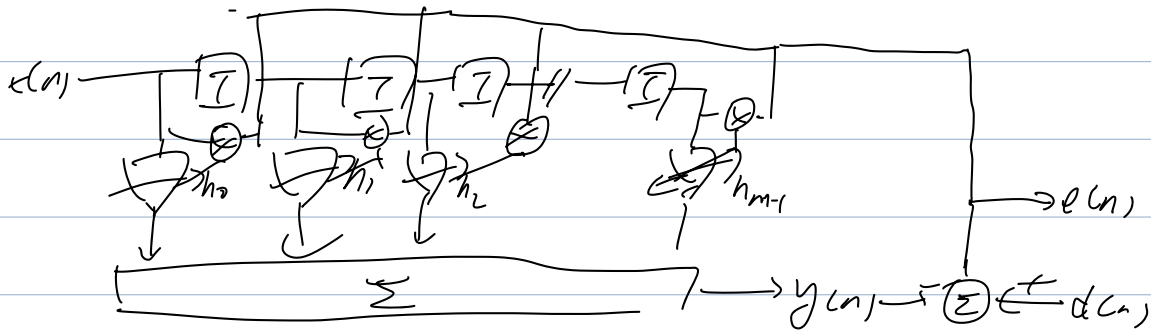
feed both soft and loud sections with almost the same level into non-linearity.

↓

100% natural.

Q2. Adaptive filters.

(a),



$$e(n) = d(n) - x(n).$$

the aim of LMS is to minimize $e(n)$,

which means it will approach to zero.

However, actually it will always not be zero, instead, it will contain any signal that is not correlated between $d(n)$ and $x(n)$.

So if we let $d(n) = \text{input} = \text{signal} + \text{noise}$

$x(n) = \text{noise}$,

then $e(n) = \text{signal}$

(b). We choose μ and ref. s.t. the filter

converges quickly but remains stable.

μ $\left\{ \begin{array}{l} \text{too large} : \text{the filter diverges. (unstable, highly oscillatory)} \\ \text{too small} : \text{it hardly removes the noise. converges slowly} \end{array} \right.$

ref. $\rightarrow$ badly scaled $(\hat{I})$: hardly removes the noise
or converges slowly.
these two should be matched.

(c).

